Minimization of Quadratic Functions under Box and Spherical Constraints

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Project Track, 29 (P) is the convex quadratic optimization problem defined on the intersection of a box and an Euclidean ball, i.e.,

$$\min \left\{ \boldsymbol{x}^T Q \boldsymbol{x} + \boldsymbol{q}^T \boldsymbol{x} : \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u},\ \|\boldsymbol{x} - \boldsymbol{c}\| \leq r \right\}$$

where $x \in \mathbb{R}^n$, q, l, u and c are fixed vectors of $\mathbb{R}^n$, r is a fixed positive real number, and Q is an $n \times n$ Positive Semidefinite matrix.

(A1) is a dual approach, cf. constrained minimization slides, with appropriate choices of the constraints to be dualized, where the Lagrangian Dual is solved by an algorithm of the class of smoothed gradient methods.

(A2) is a general-purpose solver applied to an appropriate formulation of the problem.

No off-the-shelf solvers allowed, save of course for (A2), not even for the solution of the Lagrangian relaxation.

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1 Introduction

1.1 Problem Statement

Let $\boldsymbol{x} \in \mathbb{R}^n$. We consider the following constrained minimization problem:

$$\text{(P)} \quad \min_{\boldsymbol{x} \in \mathbb{R}^n} \left\{ \boldsymbol{x}^T Q \boldsymbol{x} + \boldsymbol{q}^T \boldsymbol{x} \;\middle|\; \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u},\ \|\boldsymbol{x} - \boldsymbol{c}\|^2 \leq r^2 \right\} \tag{1}$$

Where:

- $Q \in \mathbb{R}^{n \times n}$ is a symmetric positive semi-definite matrix ($Q \succeq 0$), ensuring the convexity of the objective function.
- $\boldsymbol{q} \in \mathbb{R}^n$ is the linear coefficient vector.
- $\boldsymbol{l}, \boldsymbol{u} \in \mathbb{R}^n$ represent the lower and upper bounds of the box constraints.
- $\boldsymbol{c} \in \mathbb{R}^n$ is the center of the spherical constraint, and $r > 0$ is its radius.

1.2 First observations

First of all we notice that (1) is convex. The objective function is convex due to Q being semi-definite, and the domain is convex being an intersection of two other convex sets, a sphere and a box.

The constraints function are:

$$\begin{cases} h_l(\boldsymbol{x}) = \boldsymbol{l} - \boldsymbol{x} \leq 0 \\ h_u(\boldsymbol{x}) = \boldsymbol{x} - \boldsymbol{u} \leq 0 \\ g(\boldsymbol{x}) = \boldsymbol{x}^T \boldsymbol{x} - 2\boldsymbol{c}^T \boldsymbol{x} + \boldsymbol{c}^T \boldsymbol{c} - r^2 \leq 0 \end{cases} \tag{2}$$

We aim to show that strong duality holds via Slater's condition. The latter requires the constraint functions to be convex (which holds in our case) and the existence of a point $\boldsymbol{x}_0$ for which the constraints are satisfied strictly.

This is evidently true if the intersection of the box and the sphere has a non-empty interior. We prove that either this is true, or the intersection is naive (containing at most one point).

Suppose the intersection contains two distinct points $\boldsymbol{x}$ and $\boldsymbol{y}$. By convexity, the segment connecting them lies entirely within the intersection. We observe that any point strictly inside this segment must lie in the interior of the sphere. Indeed:

$$\|\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y} - \boldsymbol{c}\|^2 \leq \lambda \|\boldsymbol{x} - \boldsymbol{c}\|^2 + (1-\lambda)\|\boldsymbol{y} - \boldsymbol{c}\|^2 \leq r^2$$

with second equality holding if and only if both $\boldsymbol{x}$ and $\boldsymbol{y}$ are on the border of the sphere. And first equality holding if and only if $\boldsymbol{x} - \boldsymbol{c}$ and $\boldsymbol{y} - \boldsymbol{c}$ are multiple of one another. So, imposing both would mean either $\boldsymbol{x} = \boldsymbol{y}$ (which is a contradiction), or $\boldsymbol{x} = 2\boldsymbol{c} - \boldsymbol{y}$ which would obviously imply having points on the interior.

Now given a point in the interior of the sphere, there exists a direction $\boldsymbol{d}$ along which moving would satisfy all box constraints strictly. Since we can move along $\boldsymbol{d}$ for an

arbitrarily small amount $\varepsilon > 0$, we can find a point lying in the interior of both the box and the sphere.

This implies that the intersection has a non-empty interior. Thus, we can conclude that Slater's condition is met in all non-trivial cases.

Remark 1.1. To check what case our domain is in, we can project the center c onto the box: $c_i^\pi = \max(l_i, \min(u_i, c_i))$. Let $d = \|c - c^\pi\|$, if $d = r$ we have only one point and the minimum is thus achieved at c^π, if $d > r$ the intersection is empty. Lastly, if $d < r$ the intersection is non-trivial.

2 Lagrangian duality

We build a partial Lagrangian by dualizing the box constraints and keeping the ball constraint explicit. Let $\alpha, \beta \in \mathbb{R}^n_{\geq 0}$ be the multipliers for $\ell - x \leq 0$ and $x - u \leq 0$. Define the ball

$$B := \{x \in \mathbb{R}^n : \ \|x - c\|^2 \leq r^2\}.$$

The partial Lagrangian is

$$\begin{aligned} L(x, \alpha, \beta) &= x^\top Q x + q^\top x + \alpha^\top(\ell - x) + \beta^\top(x - u) \\ &= x^\top Q x + (q - \alpha + \beta)^\top x + \alpha^\top \ell - \beta^\top u. \end{aligned} \tag{3}$$

The associated dual function is

$$\psi(\alpha, \beta) := \min_{x \in B} L(x, \alpha, \beta) = \alpha^\top \ell - \beta^\top u + \min_{x \in B} \Big\{ x^\top Q x + (q - \alpha + \beta)^\top x \Big\}. \tag{4}$$

The Lagrangian dual reads

$$\text{(D)} \qquad \max_{\alpha \geq 0,\, \beta \geq 0} \ \psi(\alpha, \beta). \tag{5}$$

3 Numerical Algorithms

3.1 Solving of the outer loop

We now want to end up in the hypothesis of Nesterov 2005, and for doing so we need to manipulate our equations a little bit.

Define the convex function on B

$$\widehat{\phi}(x) := x^\top Q x + q^\top x, \qquad \text{and the linear map } A : \mathbb{R}^{2n} \to \mathbb{R}^n, \ \ A(\alpha, \beta) := \alpha - \beta.$$

Then we can rewrite

$$\begin{aligned} -\psi(\alpha, \beta) &= -\alpha^\top \ell + \beta^\top u + \max_{x \in B} \Big\{ (\alpha - \beta)^\top x - \widehat{\phi}(x) \Big\} \\ &= \underbrace{(-\alpha^\top \ell + \beta^\top u)}_{=: \widehat{f}(\alpha, \beta)} + \max_{x \in B} \Big\{ \langle A(\alpha, \beta), x \rangle - \widehat{\phi}(x) \Big\}. \end{aligned} \tag{6}$$

Hence the dual maximization (6) is equivalent to the convex minimization

$$\min_{(\alpha, \beta) \in \mathbb{R}^{2n}_{\geq 0}} F(\alpha, \beta), \qquad F(\alpha, \beta) := -\psi(\alpha, \beta). \tag{7}$$

This is exactly the "$\widehat{f} + \max$" structure required by Nesterov's smoothing framework.

3.1.1 Bounding

In order to apply Nesterov's method, the feasible set of the dual variables must be compact. We therefore derive explicit upper bounds for the optimal multipliers α^*, β^* associated with the box constraints $\ell \leq x \leq u$.

Let f^* denote the optimal primal value. Then, thanks to strong duality, any optimal dual solution (α^*, β^*) satisfies

$$\psi(\alpha^*, \beta^*) = f^*.$$

Assume there exists a point $\tilde{x} \in B$ and a margin $\rho > 0$ such that

$$\ell + \rho\mathbf{1} \leq \tilde{x} \leq u - \rho\mathbf{1}.$$

Let (α^*, β^*) be an optimal dual solution. By definition of the dual function, for any $x \in B$ we have

$$\psi(\alpha^*, \beta^*) = \min_{y \in B} \Big(f(y) + \alpha^{*\top}(\ell - y) + \beta^{*\top}(y - u) \Big) \leq f(x) + \alpha^{*\top}(\ell - x) + \beta^{*\top}(x - u).$$

Using strong duality and choosing $x = \tilde{x}$ gives us

$$f^* \leq f(\tilde{x}) + \alpha^{*\top}(\ell - \tilde{x}) + \beta^{*\top}(\tilde{x} - u).$$

Since $\alpha^*, \beta^* \geq 0$ and $\ell - \tilde{x} \leq -\rho\mathbf{1}$, $\tilde{x} - u \leq -\rho\mathbf{1}$, we obtain

$$\alpha^{*\top}(\ell - \tilde{x}) \leq -\rho\|\alpha^*\|_1, \qquad \beta^{*\top}(\tilde{x} - u) \leq -\rho\|\beta^*\|_1,$$

hence

$$\|\alpha^*\|_1 + \|\beta^*\|_1 \leq \frac{f(\tilde{x}) - f^*}{\rho} \leq \frac{f(\tilde{x}) - f_{\text{lb}}}{\rho} = K$$

for any lower bound $f_{\text{lb}} \leq f^*$. Consequently, componentwise,

$$0 \leq \alpha_i^* \leq K, \qquad 0 \leq \beta_i^* \leq K, \quad i = 1, \ldots, n.$$

We therefore restrict the dual iterates to the compact set $Q_1 = [0, \bar{\alpha}] \times [0, \bar{\beta}]$ with $\bar{\alpha}_i = \bar{\beta}_i = (f(\tilde{x}) - f_{\text{lb}})/\rho$.

We compute ρ by bisection on $[0, \rho_{\max}]$, where

$$\rho_{\max} = \tfrac{1}{2} \min_i (u_i - \ell_i).$$

At each iteration we test feasibility by projecting c onto the shrunken box $[\ell + \rho, u - \rho]$:

$$x(\rho) = \Pi_{[\ell+\rho, u-\rho]}(c) = \min(\max(c, \ell + \rho\mathbf{1}), u - \rho\mathbf{1}),$$

and checking whether $\|x(\rho) - c\|_2 \leq r$.

We don't need precision, we stop as soon as we find a ρ which is suitable and different from 0.

3.1.2 Smoothing

We now smooth the max-term in (6).

Let $Q_2 := B$ and choose the prox-function

$$d_2(x) := \tfrac{1}{2}\|x - c\|_2^2. \tag{8}$$

Then d_2 is 1-strongly convex w.r.t. $\|\cdot\|_2$ and

$$D_2 := \max_{x \in Q_2} d_2(x) = \tfrac{1}{2} r^2.$$

Let $Q_1 := [0, K\mathbf{1}] \times [0, K\mathbf{1}]$ and use prox-function on $\mathbb{R}^{2n}$,

$$d_1(\lambda) := \tfrac{1}{2}\|\lambda - \lambda^0\|_2^2, \quad \lambda := (\alpha, \beta) \in Q_1,$$

with $\lambda^0 = \frac{1}{2}(K\mathbf{1}, K\mathbf{1}) \in Q_1$. In this way it holds $D_1 = \frac{1}{8}\left(\|K\mathbf{1}|^2 + \|K\mathbf{1}\|^2\right) = \frac{1}{4} n K^2$

For $\mu > 0$ define the smoothed approximation of the max-term

$$g_\mu(\lambda) := \max_{x \in Q_2} \Big\{ \langle A\lambda, x\rangle - \widehat{\phi}(x) - \mu d_2(x) \Big\}. \tag{9}$$

Then the smoothed dual objective is

$$F_\mu(\lambda) := \widehat{f}(\lambda) + g_\mu(\lambda). \tag{10}$$

Let $x_\mu(\lambda)$ denote the unique maximizer in the definition of $g_\mu(\lambda)$. Still from Nesterov method we have:

$$\nabla g_\mu(\lambda) = A^\top x_\mu(\lambda), \qquad \nabla F_\mu(\lambda) = \nabla \widehat{f}(\lambda) + A^\top x_\mu(\lambda). \tag{11}$$

Here $\widehat{f}(\alpha, \beta) = -\alpha^\top \ell + \beta^\top u$, hence

$$\nabla_\alpha \widehat{f} = -\ell, \qquad \nabla_\beta \widehat{f} = u.$$

Since $A(\alpha, \beta) = \alpha - \beta$, we also have

$$A^\top x = (x,\, -x) \in \mathbb{R}^{2n}.$$

Therefore the gradient has the explicit form

$$\nabla_\alpha F_\mu(\alpha, \beta) = -\ell + x_\mu(\alpha, \beta), \qquad \nabla_\beta F_\mu(\alpha, \beta) = u - x_\mu(\alpha, \beta) \tag{12}$$

3.1.3 Lipschitz constant of the smoothed gradient

Under Euclidean norms, the gradient of F_μ is Lipschitz with constant

$$L_\mu = \frac{\|A\|_{1,2}^2}{\mu \sigma_2}, \qquad \sigma_2 = 1, \tag{13}$$

where (with Euclidean norms on both spaces)

$$\|A\|_{1,2} := \max_{\|\lambda\|_2 = 1} \|A\lambda\|_2, \qquad \lambda = (\alpha, \beta) \in \mathbb{R}^{2n}.$$

In our case,

$$A = \begin{bmatrix} I & -I \end{bmatrix} \in \mathbb{R}^{n\times 2n}, \qquad A(\alpha,\beta) = \alpha - \beta.$$

For any (α, β),

$$\|A(\alpha,\beta)\|_2 = \|\alpha-\beta\|_2 \le \|\alpha\|_2 + \|\beta\|_2 \le \sqrt{2}\sqrt{\|\alpha\|_2^2 + \|\beta\|_2^2} = \sqrt{2}\,\|(\alpha,\beta)\|_2,$$

so $\|A\|_{1,2} \le \sqrt{2}$.

This bound is tight: take any $v \in \mathbb{R}^n$ with $\|v\|_2 = 1$ and set $\alpha = \frac{1}{\sqrt{2}}v$, $\beta = -\frac{1}{\sqrt{2}}v$. Then $\|(\alpha,\beta)\|_2 = 1$ and

$$\|A(\alpha,\beta)\|_2 = \|\alpha-\beta\|_2 = \|\sqrt{2}\,v\|_2 = \sqrt{2}.$$

Therefore

$$\|A\|_{1,2} = \sqrt{2} \qquad \Longrightarrow \qquad L_\mu = \frac{2}{\mu}$$

(We ignore the $\widehat{f}$ term since it is linear and contributes 0 to the Lipschitz constant.)

3.1.4 Iterative method

We minimize F_μ on the bounded set Q_1 using Nesterov's method. Let Π_{Q_1} denote the projection onto Q_1 (componentwise clipping onto $[0, K\mathbf{1}]$).

Initialize $x_0 = \lambda^0 \in Q_1$. For $k = 0, 1, 2, \ldots$:

1. Compute $\nabla F_\mu(x_k)$ via (12) (inner loop provides $x_\mu(x_k)$).
2. The point y_k is defined as the minimizer of the local quadratic model of the function at x_k:
$$y_k = \arg\min_{y\in Q_1} \left\{ \langle \nabla F_\mu(x_k), y - x_k\rangle + \frac{L_\mu}{2}\|y - x_k\|^2 \right\}.$$
3. The point z_k minimizes the accumulated estimate sequence:
$$z_k = \arg\min_{\lambda\in Q_1} \left\{ \frac{L_\mu}{\sigma_1} d_1(\lambda) + \sum_{i=0}^{k} a_i \Big(F_\mu(x_i) + \langle \nabla F_\mu(x_i), \lambda - x_i\rangle \Big) \right\},$$
with $\sigma_1 = 1$ and $a_i = \frac{i+1}{2}$.
4. We now compute the new x_{k+1}:
$$\tau_k = \frac{2}{k+3}, \qquad x_{k+1} = \tau_k z_k + (1-\tau_k) y_k.$$

The auxiliary problems for y_k and z_k can be solved explicitly by reducing them to projections onto the box Q_1.

Computation of y_k By definition, y_k minimizes the local quadratic model:

$$y_k = \arg\min_{y \in Q_1} \left\{ \langle \nabla F_\mu(x_k), y - x_k \rangle + \frac{L_\mu}{2} \|y - x_k\|^2 \right\}.$$

Ignoring constant terms independent of y, this is equivalent to:

$$\begin{aligned} y_k &= \arg\min_{y \in Q_1} \left\{ \langle \nabla F_\mu(x_k), y \rangle + \frac{L_\mu}{2} \|y\|^2 - L_\mu \langle x_k, y \rangle \right\} \\ &= \arg\min_{y \in Q_1} \left\{ \frac{L_\mu}{2} \left(\|y\|^2 - 2\langle x_k - \frac{1}{L_\mu} \nabla F_\mu(x_k), y \rangle \right) \right\} \\ &= \arg\min_{y \in Q_1} \left\{ \frac{L_\mu}{2} \left\| y - \left(x_k - \frac{1}{L_\mu} \nabla F_\mu(x_k) \right) \right\|^2 \right\}. \end{aligned}$$

This is precisely the definition of the Euclidean projection of the point $x_k - \frac{1}{L_\mu} \nabla F_\mu(x_k)$ onto the set Q_1. Thus:

$$y_k = \Pi_{Q_1} \left(x_k - \frac{1}{L_\mu} \nabla F_\mu(x_k) \right). \tag{14}$$

Computation of z_k The point z_k is defined as the minimizer of the estimate sequence:

$$z_k = \arg\min_{\lambda \in Q_1} \left\{ \frac{L_\mu}{2} \|\lambda - \lambda^0\|^2 + \sum_{i=0}^{k} a_i \langle \nabla F_\mu(x_i), \lambda - x_i \rangle \right\}.$$

Let $S_k = \sum_{i=0}^{k} a_i \nabla F_\mu(x_i)$. Dropping terms independent of λ, the objective becomes:

$$\begin{aligned} z_k &= \arg\min_{\lambda \in Q_1} \left\{ \frac{L_\mu}{2} \|\lambda\|^2 - L_\mu \langle \lambda^0, \lambda \rangle + \langle S_k, \lambda \rangle \right\} \\ &= \arg\min_{\lambda \in Q_1} \left\{ \frac{L_\mu}{2} \left(\|\lambda\|^2 - 2\langle \lambda^0 - \frac{1}{L_\mu} S_k, \lambda \rangle \right) \right\} \\ &= \arg\min_{\lambda \in Q_1} \left\{ \frac{L_\mu}{2} \left\| \lambda - \left(\lambda^0 - \frac{1}{L_\mu} S_k \right) \right\|^2 \right\}. \end{aligned}$$

This corresponds to projecting the shifted center onto Q_1:

$$z_k = \Pi_{Q_1} \left(\lambda^0 - \frac{1}{L_\mu} \sum_{i=0}^{k} a_i \nabla F_\mu(x_i) \right). \tag{15}$$

3.1.5 Choosing the smoothing parameter

Given a target accuracy ε, a standard choice is $\mu = \varepsilon/(2D_2) = \varepsilon/r^2$, so that the smoothing error $\mu D_2 \leq \varepsilon/2$ and the remaining $\varepsilon/2$ is achieved by optimizing F_μ to that accuracy. Since $L_\mu = 2/\mu$, this implies $L_\mu = 2r^2/\varepsilon$.

3.1.6 Complexity

Using the complexity from the Nesterov's method and substituting our values, we obtain a complexity:

$$\begin{aligned}
&4\|A\|_{1,2}\sqrt{\frac{D_1 D_2}{\sigma_1\sigma_2}}\cdot\frac{1}{\varepsilon}+2\sqrt{\frac{MD_1}{\sigma_1\varepsilon}}= \\
&4\sqrt{2}\sqrt{\frac{(\frac{1}{4}nK^2)(\frac{1}{2}r^2)}{1\cdot 1}}\cdot\frac{1}{\varepsilon}+0= \\
&4\sqrt{2}\left(\frac{Kr\sqrt{n}}{2\sqrt{2}}\right)\cdot\frac{1}{\varepsilon}= \\
&\frac{2Kr\sqrt{n}}{\varepsilon}
\end{aligned} \tag{16}$$

where M is the Lipschitz constant of $\hat{f}$, which is 0 in our case.

3.2 Solving of the inner loop

The efficiency of the whole method relies critically on the fast solution of the inner maximization problem. Remember that for a fixed dual variable $\lambda = (\alpha, \beta)$, we must calculate:

$$x_\mu(\lambda) = \arg\max_{x\in B}\left\{\langle A\lambda, x\rangle - \hat{\phi}(x) - \mu d_2(x)\right\},$$

Rearranging the terms, this maximization is equivalent to the minimization of a strictly convex quadratic function over the Euclidean ball:

$$x_\mu(\lambda) = \arg\min_{\|x-c\|\le r}\left\{\frac{1}{2}x^\top\tilde{Q}x + \tilde{q}^\top x\right\}, \tag{17}$$

where the Hessian matrix and linear term are given by:

$$\tilde{Q} = 2Q + \mu I, \quad \tilde{q} = q - A^\top\lambda - \mu c.$$

Since $Q \succeq 0$ and $\mu > 0$, the matrix $\tilde{Q}$ is strictly positive definite.

The KKT conditions for problem (17) state that x^* is a solution if and only if there exists a Lagrange multiplier $\gamma \ge 0$ such that:

$$(\tilde{Q} + 2\gamma I)x^* + \tilde{q} - 2\gamma c = 0, \tag{18}$$

$$\gamma(\|x^* - c\| - r) = 0, \tag{19}$$

$$\|x^* - c\| \le r. \tag{20}$$

From (18), the optimal solution $x^*(\gamma)$ can be expressed explicitly as a function of the multiplier γ:

$$x^*(\gamma) = (\tilde{Q} + 2\gamma I)^{-1}(2\gamma c - \tilde{q}). \tag{21}$$

Due to complementary slackness (19), two cases arise:

1. $\gamma = 0$: $x_\mu(0)$ is strictly inside the ball.
2. $\gamma > 0$: $x_\mu(0)$ is on the boundary. In this case we must find the unique $\gamma > 0$ satisfying:
$$\|x^*(\gamma) - c\| = r.$$

Let's observe that $\tilde{Q}$ is constant throughout the entire algorithm. Thus, we can diagonalize $\tilde{Q}$ once at the start of the algorithm:

$$\tilde{Q} = V\Lambda V^\top, \quad \text{where } \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

Let $\hat{u} = V^\top(\tilde{Q}c + \tilde{q})$. Substituting Λ into (21), the condition $\|x^*(\gamma) - c\|^2 = r^2$ transforms into the following equation:

$$\begin{aligned}
\phi(\gamma) :&= \|x^*(\gamma) - c\|^2 - r^2 \\
&= \|(\tilde{Q} + 2\gamma I)^{-1}(2\gamma c - \tilde{q} - (\tilde{Q} + 2\gamma I)c)\|^2 - r^2 \\
&= \|-(\tilde{Q} + 2\gamma I)^{-1}(\tilde{Q}c + \tilde{q})\|^2 - r^2 \\
&= \sum_{i=1}^{n} \frac{\hat{u}_i^2}{(\lambda_i + 2\gamma)^2} - r^2 = 0
\end{aligned} \tag{22}$$

The function $\phi(\gamma)$ is strictly decreasing and convex on $(0, \infty)$. We solve (22) using Newton's method, which converges quadratically. The required derivatives are computed in $O(n)$ time:

$$\phi'(\gamma) = -4 \sum_{i=1}^{n} \frac{\hat{u}_i^2}{(\lambda_i + 2\gamma)^3}$$

Remark 3.1. In practice we will diagonalize only $2Q$, and then add to its eigenvalues μ.

3.2.1 Complexity

The dominant computational cost is the diagonalization of $\tilde{Q}$, requiring $O(n^3)$ operations. This is performed only once. Inside the loop, in the case of $\gamma = 0$ we requires $O(n^2)$ for matrix-vector multiplication. On the other hand, if $\gamma > 0$, finding γ takes negligible $O(k \cdot n)$ time (where k is the small number of Newton iterations), followed by an $O(n^2)$ reconstruction of x^*.

3.3 Implementation improvements

Adaptive ε The first thing that could be noticed when performing numerical tests, is that most of the iteration were used to move away from a zone of points where the projected gradient was high in norm. This led me to the idea of iteratively perform the algorithm described above starting with a higher value of ε (like from $\varepsilon = 1$) lowering over iterations. Everytime we restart the algorithm though, we start from the previous $\lambda = (\alpha, \beta)$ so to have a warm-start here as well.

Remark 3.2. To improve the initial step even more, I decided to make the initial ε start from $\log k(Q)$, so that the more ill-conditioned the matrix is the slower we try to approach the smoothed minimum.

To decrease ε we look at when the norm of the projected gradient becomes less than $T\mu$.

Let's see it better why a proportionality between that norm and μ makes sense.
For larger values of k (iterations), we can say that $x_k \approx y_k$ which is just the projection of $x_{k-1} - \frac{1}{L_\mu}\nabla F_\mu(x_{k-1})$ on the box, so we can see this as an usage of the projected gradient with factor $\frac{1}{L_\mu} = \frac{\mu}{2} = \frac{\varepsilon}{2r^2}$. Now let's suppose that the projected gradient respects

$\|p\| \leq T\mu = T\frac{\varepsilon}{r^2}$. Then:

$$\|x_k - x_{k-1}\| \leq \frac{1}{L_\mu}\|p\| \leq \frac{T\varepsilon^2}{2r^4}$$

Now, from the Nesterov Method we know that the number of iterations to obtain an ε-solution is, as computed in (16), $k_{\max} = \frac{2Kr\sqrt{n}}{\varepsilon}$. So:

$$\|x^* - x_k\| \leq \|x^* - x_{k_{\max}}\| + \|x_{k_{\max}} - x_k\| \leq \varepsilon + \frac{2Kr\sqrt{n}}{\varepsilon}\frac{T\varepsilon^2}{2r^4} = \varepsilon\left(\frac{T\sqrt{n}K}{r^3} + 1\right)$$

Let's notice two things:

- The goal of the mid stage is not to find the most precise solution, but to prepare for the next stages
- this argument is heuristic: the projected gradient is not monotonic
- the upper bound is very loose, indeed the constant $\frac{\sqrt{n}K}{r^3}$ can be quite big, to be theoretically rigorous we would need a very small T to be sure to be close enough to $x*$. However in practice even using a larger T is safe.

indeed, using $T \approx 1$ works magnificently.

All of this was incredibly beneficial, making the algorithm more efficient and effective than the pre-implemented matlab function fmincon.

Remark 3.3. Since the argument above is heuristical, another safer check was added for decreasing the ε. We average the projected gradient norm over a window of fixed size and see if we have a decreasing trend or not, if not, we decrease the ε (this with a patience).

Warm-start for Newton inner method Another important improvement to the time efficiency was the usage of a warm-start for the inner newton-method, using the solution ov the previous outer iterations as the starting point.

Remark 3.4. We could have the doubt for the Newton method for solving (22) to reach 0 in some of the denominators. However this won't ever be a problem. Indeed we have the constraint $\gamma \geq 0$ and all the λ_i are strictly positive thanks to $\tilde{Q}$ being strictly positive definite.

4 Numerical Results

4.1 Experimental Setup

The algorithms were implemented in MATLAB. All tests were performed on a standard workstation. The problem data was generated as follows: $l_i = -1.5, u_i = 1.5$ for all i, $r = 1.0$, and c, q with entries drawn from a standard normal distribution. For the full rank case, $Q = AA^T$ with $A \in \mathbb{R}^{n \times n}$. For the low rank case, $Q = AA^T$ with $A \in \mathbb{R}^{n \times 0.2n}$.

The specific hyperparameters used for the proposed method and the reference `fmincon` solver are detailed in Table 1.

Table 1: Algorithm Hyperparameters

Proposed method		**fmincon (Interior-Point)**	
Parameter	*Value*	*Parameter*	*Value*
Target Accuracy (ϵ)	10^{-15}	Constraint Tol.	10^{-12}
PG Tolerance	10^{-15}	Optimality Tol.	10^{-15}
Inner Loop ϕ Tol.	10^{-15}	Step Tol.	10^{-10}
Inner Loop Step Tol.	10^{-15}	Max Iterations	5000
ρ Bisection Iters	1e6	Max Func Evals	2×10^6
μ Init Factor	$\frac{k(Q)}{\epsilon}$		
μ Decay Factor	10		
Min Iter per Stage	10		
Optimality Factor	0.5		
Stagnation Start Iter	1000		
Stagnation Tol.	0.001		
Window Size	30		
Patience	30		

- **Target Accuracy ($\epsilon = 10^{-15}$):** Defines the desired precision for the final solution.
- **Projected Gradient Tolerance (`tol_pg` $= 10^{-15}$):** The global stopping criterion. The algorithm terminates when the current smoothing parameter is $\mu_{\min}$ and the norm of the projected gradient drops below this threshold.
- **Inner Loop Tolerances (`tol_inner_phi`, `tol_inner_step` $= 10^{-15}$):** Control the precision of the Newton method.
- **Bounding Iterations (`rho_bisect_it` $= 1e6$):** The number of bisection iterations used to find the margin ρ and subsequently the bound K for the dual variables box constraints.
- **Smoothing Schedule (`mu_init_factor` $= \frac{k(Q)}{\epsilon}$, `mu_decay` $= 10$):** Control the continuation strategy. The algorithm starts with a large $\mu_0 = \mu_{\min} \cdot$`mu_init_factor` to exploit a smoother landscape and faster convergence. Upon completing a stage, μ is reduced by dividing it by `mu_decay`.
- **Stage Switching Condition (`optimality_factor` $= 0.5$):** The primary condition for reducing μ. A stage is considered complete when the norm of the projected gradient satisfies $\|p\| \leq 0.5\mu$.
- **Stagnation Detection (`window_size`, `patience`, `stag_tol`):** A heuristic mechanism to force a μ-update if the algorithm struggles to reach the optimality condition.

If the relative improvement of the projected gradient norm (averaged over a window of 30 iterations) is less than 0.001 for 30 consecutive checks (`patience`), the algorithm forces a transition to the next μ, provided at least 1000 iterations (`after_it`) have passed.

Remark 4.1. The apparently high values for some of the hyper-parameters of fmincon have been used due to the method being unable to output feasible points with lower values of those parameters.

4.2 Performance Comparison

The comparison between the proposed method and `fmincon` is summarized in tables 2 and 3. The Viol columns represent the violation of box and ball constraints. As we can see the proposed method violates them with in the order of machine precision.

n	$\kappa(\mathbf{Q})$	**Time (s)**		**Accuracy**		**Viol. Dual**		**Viol. fmin**	
		Dual	fmin	$f_{dual} - f_{fmin}$	Rel. Diff $\mathbf{x}$	Box	Ball	Box	Ball
500	3.8×10^5	**0.056**	16.32	-6.4×10^{-8}	7.7×10^{-9}	0	2.2×10^{-16}	0	0
1000	1.5×10^7	**0.243**	145.71	-2.3×10^{-6}	1.6×10^{-7}	0	2.2×10^{-16}	0	0
3000	2.3×10^8	**4.158**	$*$	-8.9×10^4	1.5×10^{-1}	0	2.2×10^{-16}	0	3.5×10^{-1}

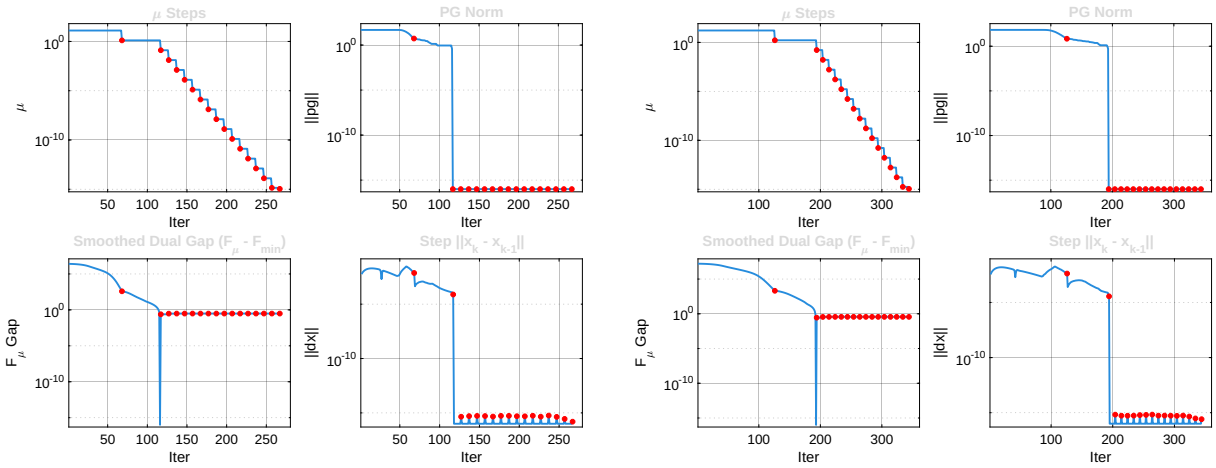
Table 2: Results for full rank matrices.

n	**Time (s)**		**Accuracy**		**Viol. Dual**		**Viol. fmin**	
	Dual	fmin	$f_{dual} - f_{fmin}$	Rel. Diff $\mathbf{x}$	Box	Ball	Box	Ball
500	**0.042**	7.44	-2.6×10^{-9}	1.1×10^{-9}	0	2.2×10^{-16}	0	0
1000	**0.136**	36.33	-8.0×10^{-6}	7.4×10^{-8}	0	4.4×10^{-16}	0	0
3000	**1.557**	$*$	-1.1×10^1	1.3×10^{-4}	0	2.2×10^{-16}	0	0

Table 3: Results for low rank matrices (rank $\approx 0.2n$).

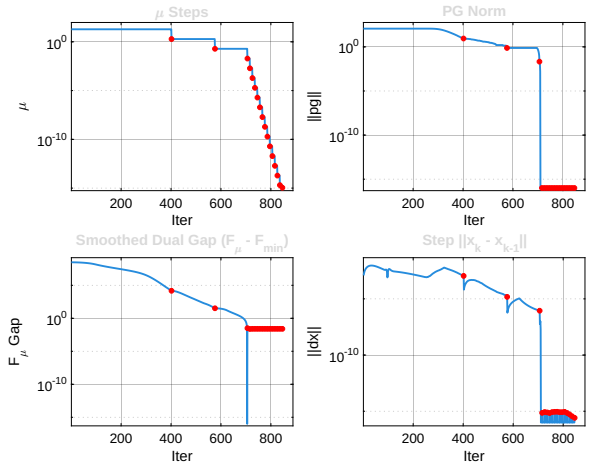
* means that the time limit of 10min has been reached.

4.3 Graphs



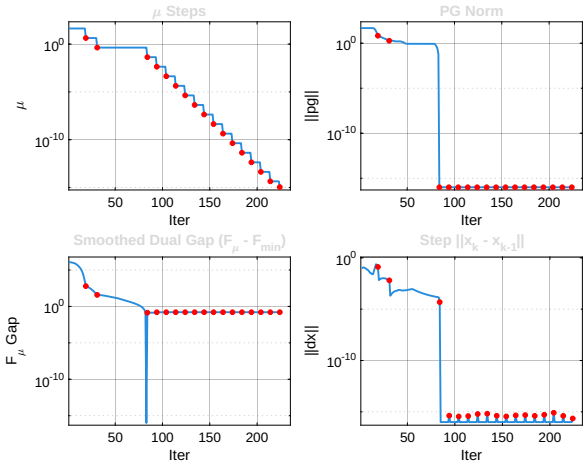
(a) Dimension $n = 500$

(b) Dimension $n = 1000$

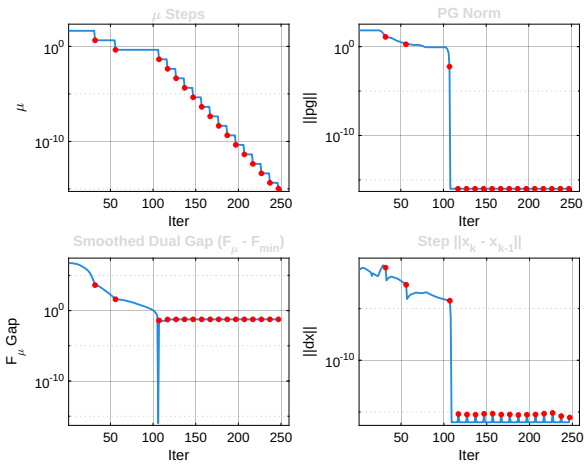


(c) Dimension $n = 3000$

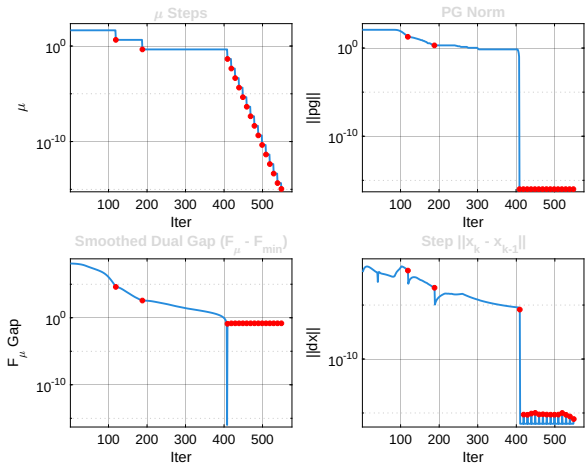
Figure 1: Performance analysis for full rank matrices. Each panel shows (top-left to bottom-right): smoothing parameter μ, PG norm, dual gap $F_\mu - F_{\min}$, and step size $\|x_k - x_{k-1}\|$.



(a) Dimension $n = 500$



(b) Dimension $n = 1000$



(c) Dimension $n = 3000$

Figure 2: Performance analysis for low rank matrices. Similar to the full rank case, the plots illustrate the convergence behaviors.

References

Nesterov, Yu (2005). "Smooth minimization of non-smooth functions". In: *Mathematical programming* 103.1, pp. 127–152.